Shafayet Sadik Sowad

Date of birth: 07/06/2002 | Gender: Male | Nationality: Bangladeshi | Phone: (+880) 1780622406 (Mobile) | Email address: shafayetsadiksowad@gmail.com | Website: https://www.linkedin.com/in/shafayet-sadik-sowad-ab61a2226/ | Address: Sector 15, Uttara, 1230, Dhaka, Bangladesh (Home)

Studies Applied For

Erasmus Mundus Joint Master (EMJM) in Intelligent Field Robotic Systems (IFROS)

Education & Training

BSc in Electrical and Electronic Engineering | Islamic University of Technology | 07/06/2021 - 25/10/2025 | Gazipur, Bangladesh

Final grade: 3.64 | **Level in EQF:** 6 | **Number of credits:** 183.50 | **Thesis:** Triplet-Loss-Driven Contrastive Autoencoding for Anomaly-Based Handover Prediction Using Drive-Test Data in Dynamic Cellular Environment

Work experience

Systems Engineer | ANTS Aerial Systems | 11/10/2025 - Current | Dhaka, Bangladesh

- Developed light show drone systems with synchronized flight control and formation execution, including a custom light show designer.
- Developed a drone surveillance system with real-time monitoring capabilities.
- Planned and led agricultural drone operations and field testing for precision mapping and autonomous flight missions.
- Performed agricultural drone mapping over approximately 100 sq. km for survey and planning applications.
- Developed drone-based real estate window view simulation workflows.
- Developed a custom path planner for drone-based window view simulation.
- Contributed to circuit schematic design and PCB development.
- Worked on drone thermography for thermal inspection, data acquisition, and analysis.
- Worked on thermal image processing using OpenCV for temperature mapping and anomaly detection.
- Developed a drone-based system for solar shading analysis.
- Conducted R&D on custom payload drones.
- Conducted R&D on GPS-denied drone systems.

R&D Engineer Intern | Cybernetics Hi-Tech Solutions (Pvt.) Ltd. | 01/06/2024 - 10/04/2026 | Dhaka, Bangladesh

- Studied energy metering ICs, analyzed register structures and reference code, and prepared technical documentation.
- Selected motors and rotary actuators based on mechanical and electrical requirements, including speed, torque, gearbox selection, and datasheet analysis.
- Worked on drone communication systems, studying SBUS, UART, USB, and MAVLink-based data transfer.
- Configured LiDAR, optical sensors, and Pixhawk parameters for sensing and flight-control integration.
- Contributed to BMS circuit design and schematic development.
- Assisted in PCB schematic and Eagle library development, with exposure to commercial PCB design practices such as filtering, redundancy, and signal integrity.
- Reviewed research papers for AGV project to support technical understanding and system design decisions.

Projects

MARS Rover

Autonomous SLAM Drone

Underwater ROV Design

Real-Time IoT based Monitoring, controlling and automated System

Lipo 6s Battery Management System (BMS).

SAP-I 8-bit Computer with ICs, Registers, and Flip-flops

Built basic computer with hardware components using Proteus for simulation and custom PCBs.

Alarm Clock with 8051 Microcontroller Assembly Coding

Real-time countdown, keypad interface, LCD, 7 segment displays and puzzle-solving functionality.

Design of 8:1 Analog Multiplexer Circuit

Designed and investigated temperature sensor circuit for smart food packaging.

Remote Battery level Monitoring and tracking system with Blynk IoT and GPS

Solar panel cleaning System with Pump Control using SIM module

SIMULINK Model for a Prosthetic Leg Gait Cycle.

Developed a human leg gait cycle model using neural networks and RBF function in SIMULINK.

Dynamic Voting System using Google App script.

Capacitor Discharge Flyback Transformer for Pulsed High voltage Generation.

Publications

Drive-Test-Based LTE Handover Dataset for Cellular Mobility Studies in Urban Bangladesh

Data in Brief

Reducing Handover and Ping-Pong Events in LTE Networks via Q-Learning and Subtractive Clustering

2025 8th International Seminar on Research of Information Technology and Intelligent Systems

Adaptive Handover Optimization in LTE: Comparative Evaluation of Fuzzy Logic and DQN

2025 IEEE Asia Pacific Conference on Wireless and Mobile

Handover Optimization in LTE Networks Using Contextual Bandit Reinforcement Learning and Real-World Data

2025 International Conference on Computer Engineering, Network and Intelligent Multimedia

Triplet-Loss-Driven Contrastive Autoencoding for Anomaly-Based Handover Prediction Using Drive-Test Data in Dynamic Cellular Environment

Paper Under Review

Language Skills

Mother tongue(s): **Bengali**

	Understanding		Speaking		Writing
	Listening	Reading	Spoken production	Spoken interaction	
English	C1	C1	C1	C1	B2

Skills

Embedded Systems and Robotics | ROS | SLAM | OpenCV | Linux | Kinematics | Visual Odometry | Gazebo | PCB Design | CAD

Coding | Python | C++ | Object Oriented Programming | Git | VS code | Machine Learning | Deep Learning | MATLAB | Assembly | LaTeX

Circuit Design and VLSI | Cadence Virtuoso | Proteus | Simulink | PSpice | Orcad | Fusion 360 | KiCad | Altium | Eagle | Fritzing | Verilog

Spatial Analysis | QGIS | Agisoft metashape

Honours and Awards

18th Place as Project Altair | University Rover Challenge 2025, USA

2nd Place as Project Aqua | Underwater Vehicle Design Challenge 2023, IIT Guwahati

Top 12 Finalist as Project Aero | International Space Drone Challenge 2024, India

6th Place as Project Altair (ABEX Task Winner) | International Rover Challenge 2024

17th Place as Project Altair | European Rover Challenge 2023, Kielce, Poland

Scholarship for Academic Excellence in HSC (2020) | Bangladesh Govt.

Scholarship for Academic Excellence in SSC (2018) | Bangladesh Govt.

Scholarship for Academic Excellence in JSC (2016) | Bangladesh Govt.

Co-curricular activities

Team Lead-Project Aero (Drone Project)

Vice Chair - IEEE RAS IUT Student Branch Chapter

Head of Operations - IUT Robotics Society

Senior Electrical Executive - Project Altair (Mars Rover)

Head of PCB - Accelerate (Inter University PCB Design Competition)

Head of Seismic Challange - Cennovation (Civil Departmental Fest)

Administrative Member - IUT Al-Fazari Interstellar Society

References

Dr. Mohammad Tawhid Kawser
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Islamic University of Technology (IUT)
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Mr. Nadim Ahmed
Assistant Professor, Department of Electrical and Electronic Engineering
Islamic University of Technology (IUT)
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